

SOC INCIDENT INVESTIGATION REPORT

Windows Failed Login Detection using Wazuh

Project	Wazuh SOC Home Lab
Incident ID	SOC-2026-001
Detection Platform	Wazuh SIEM
Endpoint	Windows 11
Agent	windows-11 (Agent ID 001)
Analyst	Syed Taif Ahmed
Environment	Authorized Security Testing Lab
Investigation Date	12 September 2026

CONFIDENTIAL • LAB / TRAINING DOCUMENT

1. Executive Summary

This investigation was carried out in a controlled SOC home lab to check whether Wazuh could detect repeated Windows authentication failures and provide enough information for an analyst to investigate the event. I intentionally entered an incorrect password three times on the Windows 11 endpoint and then logged in successfully. Wazuh collected Windows Security Event ID 4625 and generated three Level 5 alerts under Rule 60122. The following successful login helped confirm the sequence. After reviewing the account, source address, logon type and Windows status codes, I assessed the activity as authorized lab activity rather than a real compromise.

2. Objective

The objective was to validate a basic SOC monitoring workflow: generate a known authentication event, confirm that Windows records it, verify that the Wazuh agent forwards it, review the alert in the Wazuh dashboard, and make an analyst-level decision about the activity.

3. Lab Environment

Component	Details
SIEM / Monitoring	Wazuh
Wazuh Server	Ubuntu Server VM
Endpoint	Windows 11
Wazuh Agent	windows-11
Agent ID	001
Network	VMware NAT
Log Source	Windows Security Event Log
Test Type	Controlled failed-login simulation

4. Detection Details

Field	Observed Value
Windows Event ID	4625
Wazuh Rule ID	60122
Wazuh Severity	Level 5
Alert	Logon Failure - Unknown user or bad password
Target User	Taif
Logon Type	2 - Interactive
Source IP in Event	127.0.0.1
Authentication	Negotiate
Logon Process	User32
Process	C:\Windows\System32\svchost.exe
Status	0xC000006D
SubStatus	0xC000006A
Attempts	3

5. Incident Timeline

Time	Activity	Detection / Result
19:50:36	Failed Windows login	Event ID 4625 / Rule 60122
19:50:39	Failed Windows login	Event ID 4625 / Rule 60122
19:50:42	Failed Windows login	Event ID 4625 / Rule 60122
19:50:48	Successful login	Windows workstation logon

6. Investigation Findings

- 1. Repeated authentication failures:** Three failed login attempts were observed within a short period. Wazuh generated three corresponding authentication-failure alerts.
- 2. Account involved:** The event targeted the local Windows account 'Taif'. The event data identifies the target domain as TAIF.
- 3. Source of activity:** The Windows security event recorded 127.0.0.1 as the source address. This indicates the test was generated locally on the endpoint, not from an external network host.
- 4. Logon type:** Logon Type 2 indicates an interactive logon. This matches the way the test was performed on the Windows workstation.
- 5. Reason for failure:** The status and sub-status values are consistent with a bad-password authentication failure. The Wazuh rule description also states 'Unknown user or bad password'.
- 6. Outcome:** A successful login followed the three failures. In the context of the controlled lab, this confirmed that the test sequence completed as intended.

7. Key Evidence

The raw Wazuh event contains many fields and is not reproduced in full here. The following values are the fields that were most useful for the investigation.

Evidence	Observed Value
Event ID	4625
Rule	60122
Severity	Level 5
User	Taif
Logon Type	2 - Interactive
Source	127.0.0.1
Failure	Unknown user name or bad password
Status	0xC000006D
SubStatus	0xC000006A
Occurrences	3

8. Analyst Assessment

Verdict: BENIGN / AUTHORIZED LAB ACTIVITY

The alert itself is a valid detection of a failed Windows authentication attempt, so it should not be ignored in a real SOC. In this exercise, however, the analyst had direct knowledge that the attempts were intentionally generated. The localhost source address, interactive logon type, bad-password status and the successful login immediately afterward all fit the expected test pattern. No evidence from this exercise indicates an external attacker or account compromise.

9. MITRE ATT&CK Context

Source / Behavior	Technique	Name / Context	Analyst Note
-------------------	-----------	----------------	--------------

Wazuh rule metadata	T1531	Account Access Removal	The rule metadata displayed this mapping. It was not treated as proof that T1531 occurred.
Observed behavior	Windows authentication failure	Event ID 4625	Repeated failed authentication was directly observed in the Windows Security log.

10. Recommended SOC Response

- For a production environment, correlate repeated failures from the same account or source over a defined time window.
- Review the source IP, workstation name and authentication context before deciding whether an alert is malicious.
- Escalate repeated remote failures, especially when they involve privileged accounts or multiple usernames.
- Keep Windows Security auditing enabled and ensure the Wazuh agent remains connected so authentication events are collected.
- Use account lockout and strong-password controls appropriate to the organization's security policy.
- For future lab exercises, extend the investigation with a Linux SSH brute-force simulation and compare the Windows and Linux detection workflows.

11. Evidence / Screenshot Placement

Recommended screenshots for the final evidence pack:

- 01-Windows-Agent-Active.png — Wazuh dashboard showing windows-11 as active.
- 02-Windows-Event-4625.png — Windows Event Viewer showing Event ID 4625.
- 03-Wazuh-Failed-Login-Alerts.png — Wazuh Events view showing the three Rule 60122 alerts.
- 04-Wazuh-4625-Key-Evidence.png — Wazuh event details with Event ID, username, logon type, source IP and status codes.
- 05-Wazuh-Successful-Login.png — successful login alert/event showing the end of the test sequence.

Screenshots can be inserted under this section after the report is reviewed. The report intentionally uses a small set of meaningful evidence instead of pasting the full raw JSON event.

12. Conclusion

The exercise successfully demonstrated a simple end-to-end SOC workflow using Windows Security logs and Wazuh. The Windows endpoint generated Event ID 4625 after incorrect credentials were entered, the Wazuh agent collected the event, and Wazuh produced Rule 60122 alerts. The activity was investigated using the available event fields and was correctly classified as authorized lab activity. This provides a practical base for the next lab exercise, where a Linux SSH authentication scenario can be monitored in the same way.

13. Analyst Note

This document describes activity performed in a private VMware lab for learning and portfolio purposes. No third-party or public system was targeted.